The homotopy head: a travel-time field on the double cover of SE(3)

G-NTFields, `models/graph_homotopy`

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Abstract

The homotopy head is a decoder for neural time fields on SE(3). It does not decode the travel time $T(x, y)$ with a free-form MLP. Instead, a pose encoder produces a few small per-pose quantities that enter a fixed formula built on a free-space travel time, evaluated on the double cover $\mathbb{R}^3 \times S^3$. The two lifts of the goal are the two homotopy classes of paths in SE(3), and T is the minimum over them. On 30 shape–environment cells and 30 T/L/A test sets the head is significantly ahead of the metric NTFields on every set; on a Gibson scene it wins one of two cells.

1 Why a double cover

A neural time field learns $T(x, y)$, the shortest travel time between poses $x, y \in \mathrm{SE}(3)$ under a speed $S(x)$ that falls to zero near obstacles, by fitting the Eikonal equation $\|\nabla_x T(x, y)\| = 1/S(x)$ (here with the Lie gradient). A planner then descends $T(\cdot, y)$ towards the goal.

Rotations make this hard. $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$, so between two poses there are exactly two homotopy classes of paths. T is the minimum of the two class times, and where they tie (the half turn in free space, moved by obstacles) T has a ridge that a smooth network fits poorly and a gradient planner can stall on.

The double cover $\mathbb{R}^3 \times S^3$ (unit quaternions, q and $-q$ distinct) is simply connected. A path from x to y lifted from $\tilde{x}$ ends at either $\tilde{y}$ or $-\tilde{y}$: the two endpoints are the two classes. So the head learns a field $\tilde{T}$ on the cover and takes

$$T(x, y) = \min\big(\tilde{T}(\tilde{x}, \tilde{y}),\ \tilde{T}(\tilde{x}, -\tilde{y})\big). \tag{1}$$

Each sheet $\tilde{T}(\cdot, \pm\tilde{y})$ is smooth; the only ridge of T is where the sheets tie, and it is placed by the topology rather than learned.

2 Architecture

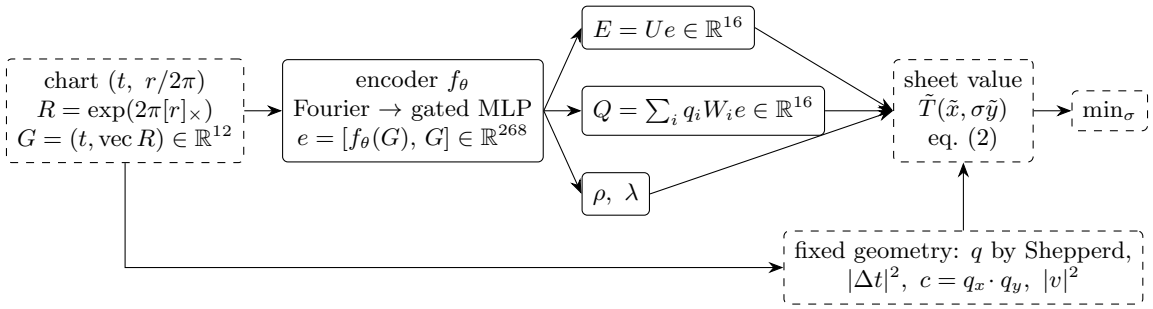


Figure 1: Data flow. Solid boxes contain learned weights; dashed boxes are fixed computations. The pose reaches the value twice: through the encoder, and directly through the fixed pair geometry.

2.1 How a pose enters

A pose is stored in the chart $(t,\ r/2\pi)$, with $t \in \mathbb{R}^3$ and r the rotation vector. It is mapped, differentiably, to $R = \exp(2\pi[r]_\times)$ and to the 12-number feature $G(x) = (t, \mathrm{vec}\, R)$. Because G uses R, everything computed from it is the same for q and $-q$. The pose is used in two separate ways:

1. **Learned path.** $G(x)$ goes through the encoder; the per-pose heads read the result.
2. **Fixed path.** The unit quaternion q is recovered from R (Shepperd's method; its sign is fixed per region, giving one lift $\tilde{x}$). The pair geometry is computed from (t, q) with no weights at all.

2.2 Encoder (learned)

The encoder is the G-Shape NTFields encoder:

$$\begin{aligned}
\gamma &= [\sin(2\pi GB),\ \cos(2\pi GB)] \in \mathbb{R}^{256}, && B \in \mathbb{R}^{12\times 128} \text{ fixed random (std 0.5 on } t\text{, 0.1 on } R),\\
u_1 &= \sin(L_1\gamma), \quad u_2 = \sin(L_2\gamma), && \text{gates,}\\
h &\leftarrow u_1 \odot \sin(Lh) + u_2 \odot (1 - \sin(Lh)) && \text{twice per block, then a gated residual,}\\
e(x) &= [\, L_{\text{out}}h,\ G(x)\,] \in \mathbb{R}^{268}.
\end{aligned}$$

There are two blocks of three 256×256 layers plus the output layer. The residual gate is a learned scalar, and each weight row is softly capped at norm 1 (Lipschitz control). The raw feature $G(x)$ is appended to the output, so the heads see the encoding and the pose itself.

2.3 Per-pose heads (learned)

All four heads are linear in $e(x)$:

Head	Formula	Weights	Behaviour under $q \to -q$
even	$E(x) = Ue$	$U \in \mathbb{R}^{16\times 268}$, no bias	unchanged
odd	$Q(\tilde{x}) = \sum_{i=0}^{3} q_i W_i e$	$W \in \mathbb{R}^{64\times 268}$ as $W_0..W_3 \in \mathbb{R}^{16\times 268}$, no bias	sign flips
cone scale	$\rho(x) = \text{softplus}(a_\rho \cdot e + b_\rho)$	$a_\rho \in \mathbb{R}^{268}$, b_ρ	unchanged
saturation	$\lambda(x) = 10^{-3} + \text{softplus}(a_\lambda \cdot e + b_\lambda)$	$a_\lambda \in \mathbb{R}^{268}$, b_λ	unchanged

Q is linear in the quaternion, which is the only place the network sees which lift a pose is on. At initialisation $a_\rho = a_\lambda = 0$ and the biases give $\rho = 0.5$, $\lambda = 0.05$ everywhere.

What is learned: 614,110 parameters, of which 592,132 are in the encoder and 21,978 in the heads (U: 4,288, W: 17,152, ρ and λ: 269 each). **What is fixed:** the Fourier matrix B, the scales $s_t = 1$ and $s_r = 2\pi$ (one full turn counts as one unit, like one unit of translation), $\lambda_{\min} = 10^{-3}$, the quaternion lift and every formula in the next subsection.

2.4 From the heads to T

Take a pair with lifts $\tilde{x}, \tilde{y}$ and a sheet sign $\sigma \in \{+1, -1\}$ ($\sigma = -1$ uses the goal's other lift $-\tilde{y}$). The value is built in five steps.

1. **Fixed geometry.** $a = |t_x - t_y|^2$, $c = q_x \cdot q_y$, $|v|^2 = 1 - c^2$ (with $q_x^* q_y = (c, v)$), and the rotation angle on the sheet $\psi_\sigma = 2\,\text{atan2}(|v|, \sigma c) \in [0, 2\pi]$. On the near sheet this is the geodesic angle θ; on the far sheet it is $2\pi - \theta$, the long way round.
2. **Free-space cone.** $T_{\text{free},\sigma} = \sqrt{a/s_t^2 + \psi_\sigma^2/s_r^2}$, the exact travel time with unit speed.
3. **Cone term.** Add $\bar{\rho}\,\bar{\lambda}\tanh(T_{\text{free},\sigma}/\bar{\lambda})$ with $\bar{\rho} = \frac{1}{2}(\rho(x) + \rho(y))$ and $\bar{\lambda}$ likewise. Close to the goal ($T_{\text{free}} \ll \bar{\lambda}$) the time is $(1 + \bar{\rho})\,T_{\text{free}}$, a slower cone; beyond $\bar{\lambda}$ the term saturates to a constant offset $\bar{\rho}\bar{\lambda}$.
4. **Even term.** Add $\|E(x) - E(y)\|^2$. It is the same on both sheets, so it is the obstacle cost that does not depend on the homotopy class.
5. **Odd term.** Add $\|Q(\tilde{x}) - \sigma Q(\tilde{y})\|^2 = \|Q(\tilde{x})\|^2 + \|Q(\tilde{y})\|^2 - 2\sigma\, Q(\tilde{x}) \cdot Q(\tilde{y})$. Its learned values are the only ones that change between the sheets; the cone term changes only through $T_{\text{free},\sigma}$.

Together,

$$\tilde{T}(\tilde{x}, \sigma\tilde{y}) = T_{\text{free},\sigma} + \bar{\rho}\,\bar{\lambda}\tanh\big(T_{\text{free},\sigma}/\bar{\lambda}\big) + \|E(x) - E(y)\|^2 + \|Q(\tilde{x}) - \sigma Q(\tilde{y})\|^2, \qquad T(x, y) = \min_{\sigma=\pm 1} \tilde{T}(\tilde{x}, \sigma\tilde{y}). \tag{2}$$

The minimum is found without gradients and only the chosen sheet is recomputed with them.

What each term does.

- *The ridge.* The sheets differ through ψ_σ (near or far way round), the cone terms, and the odd term, which contributes $-4\,Q(\tilde{x}) \cdot Q(\tilde{y})$ to $\tilde{T}(\tilde{x}, \tilde{y}) - \tilde{T}(\tilde{x}, -\tilde{y})$. Without obstacles Q can stay small and the tie sits at the half turn; with obstacles $Q(\tilde{x}) \cdot Q(\tilde{y})$ moves it, so one class can win even for rotations under a half turn.
- *The local speed.* The squared even and odd terms have zero slope at $y = x$, so near the goal only the cone is felt and the Eikonal equation forces $1 + \rho(y) \approx 1/S_d(y)$. ρ is the slowdown at a pose, and λ sets how far that slowdown carries.
- *Obstacle detours.* E and Q add time for poses that are close in free space but far apart around obstacles; they add nothing when $x = y$.

By construction, for every parameter value: $T(x, x) = 0$, $T(x, y) = T(y, x)$, $T \geq T_{\text{free}} \geq 0$, and flipping both quaternions leaves T unchanged, so T is a function on $\mathrm{SE}(3) \times \mathrm{SE}(3)$.

3 Training and planning

Loss. For a training pair with endpoint speeds, take the Lie (body-frame) gradients $g_x, g_y \in \mathbb{R}^6$ of the selected sheet with respect to each endpoint and set $N_e = S_d(e)\,\|g_e\|$, where S_d is the translational clearance speed, warped as $s^2(2-s)^2$ and clamped at 10^{-3}. The loss is the NTFields Eikonal residual

$$\mathcal{L} = \beta\,\frac{1}{n}\sum_i \Big[10^{-2}\big(\sqrt{N_{x,i}} - 1\big)^2 + 10^{-2}\big(\sqrt{N_{y,i}} - 1\big)^2\Big]\, e^{-T_i/2},$$

with the weight $e^{-T/2}$ emphasising short pairs. With `--lift min` each pair trains only its minimum sheet. AdamW (learning rate $5 \cdot 10^{-4}$, weight decay 0.5), batches of 2000 pairs from 800k, rejected epochs rolled back, 8000 epochs.

Planning uses the same forward MPPI as NTFields (50 samples, horizon 5, step 0.015, temperature 50, 200 iterations). The *dual* planner lifts every sample continuously and plans once towards each lifted goal $\pm\tilde{y}$, one plan per homotopy class, keeping the plan that converges. Success means reaching the goal within 0.01 in the SE(3) product metric with no collision on the way.

4 Results

Mean SE(3) success (%) over 1000-case test sets, 3 seeds (6 on Gibson). *Full* NTFields has the TD and normal-alignment losses on; *reduced* has them off. "Ahead" counts sets where the paired McNemar test favours the homotopy head at $p < 0.01$.

Group	sets	Homotopy	Full NTF	Reduced NTF	ahead / behind
New shapes, open room (env4)	5	98.0	90.2	88.0	5 / 0
New shapes, 1 wall, 70 window (env8)	5	96.7	89.7	87.3	5 / 0
New shapes, 1 wall, 60 window (env7)	5	82.8	73.0	71.6	5 / 0
New shapes, 1 wall, 50 window (env5)	5	55.6	51.5	51.3	5 / 0
New shapes, 2 walls, 70 windows (env9)	5	94.1	76.5	68.4	5 / 0
New shapes, 2 walls, 50 windows (env6)	5	33.6	27.7	26.8	5 / 0
T / L / A, standard	6	99.0	86.7	85.0	6 / 0
T / L / A, long / big rotation	12	99.1	81.0	78.3	12 / 0
T / L / A, tight / hard	12	97.0	66.8	62.9	12 / 0
Gibson eastville, cube	1	84.6	69.8	54.1	1 / 0
Gibson eastville, small rectangle	1	74.0	88.1	75.9	0 / 1

- The gain is in passing walls. With start and goal on the same side all models score 97–100%. Crossing both walls of env9, the homotopy head scores 87.2% against 49.0% (full NTFields).
- Narrow 50-unit windows (env5, env6) defeat both methods: crossing one wall succeeds in 11–15% of cases for the homotopy head and about 4–5% for NTFields.
- The one loss is the small rectangle in eastville, where the homotopy head collides in 24% of cases against 10% for the full NTFields.